INCH-POUND

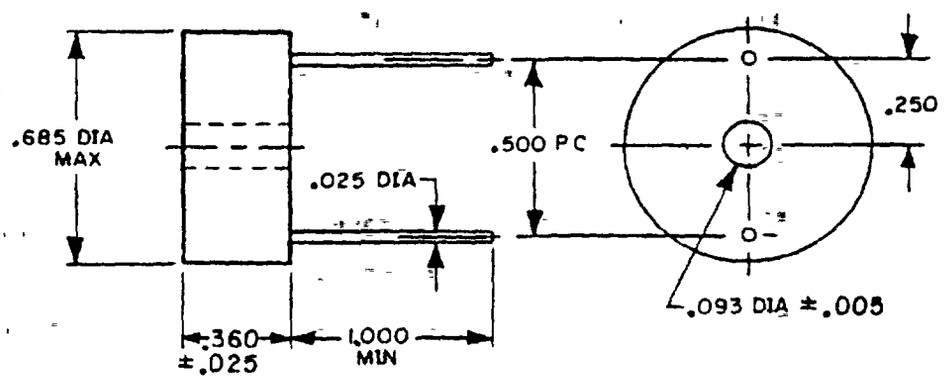
MIL-T-27/215C
19 June 1989
SUPERSEDING
MIL-T-27/215B
7 August 1980

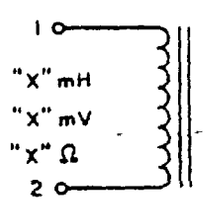
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER, AND HIGH-POWER PULSE) INDUCTORS, AUDIO FREQUENCY, TF5S20ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Ⓒ The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





WV-250 V
ALTITUDE-70,000 FT MAX
(SEE TABLE I FOR VALUE OF "X")

Inches	mm
.005	0.13
.025	0.64
.093	2.36
.250	6.35
.360	9.14
.500	12.70
.685	17.40
1.000	25.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.010.
4. Marking shall be on the side of the case opposite terminals.
5. Electrical values shall be marked as specified in table.

FIGURE 1. Dimensions and configuration.

Ⓒ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Working voltage: 250 volts.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.
Material: Epoxy.

Altitude: 70,000 feet, maximum.

Terminals: Tinned copper, 22 AWG.
Terminal length: 1 inch minimum.

Weight: 6.6 grams, maximum.

Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:
At sea level: 1,000 V rms.
At reduced barometric pressure: 300 V rms.

Electrical characteristics (see table I): Power level 1.0 watt maximum.

Temperature rise: +40°C at an ambient temperature of +90°C.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

Part number: M27/215-(dash number from table I).

(C) Supersession data: MIL-T-27/215 supersedes MIL-T-27/95(USAF).

Superseding part numbers: M27/215-14 supersedes M27/95-01.

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© TABLE I. Electrical ratings 1/

Dash No.	Inductance (mH) (1-2)	Voltage (mV) (1-2)	Self-resonant frequency (kHz) (min)	DC resistance (ohms) (1-2) (max)	Quality factor (min)	Distrib. capacitance (pF) (max)
01	0.05 ±5%	21.2	6.9	.64	45 at 100 kHz	11
02	0.10 ±5%	30.0	4.6	.9	60 at 100 kHz	12
03	0.33 ±2%	54.6	2.6	3.25	65 at 100 kHz	11
04	0.56 ±2%	20.08	1.2	.38	40 at 10 kHz	31.4
05	1.00 ±2%	94.9	1.3	8.35	35 at 50 kHz	15
06	1.00 ±2%	38.9	1.0	1.41	33 at 10 kHz	25
07	1.00 ±2%	27.0	.97	.77	45 at 10 kHz	27
08	3.30 ±1%	172.8	.72	25 9	35 at 50 kHz	15
09	3.30 ±1%	70.8	.52	4.04	33 at 10 kHz	28
10	3.30 ±1%	49.1	.5	2.22	45 at 10 kHz	31
11	10.00 ±1%	300.0	.36	88.6	32 at 50 kHz	20
12	10.00 ±1%	122.8	.26	14	45 at 10 kHz	37
13	10.00 ±1%	85.4	.23	5.75	45 at 6 kHz	48
14	10.00 ±1%	300.0	.475	90	50 at 100 kHz	11
15	15.00 ±1%	367.5	.32	165.6	32 at 50 kHz	16
16	33.00 ±1%	223.8	.132	39.4	25 at 6 kHz	44
17	33 00 ±1%	155.4	.115	21.7	45 at 6 kHz	58
18	100.00 ±1%	388.4	.068	139	25 at 6 kHz	55
19	100.00 ±1%	269.7	070	57.2	45 at 6 kHz	52
20	330.00 ±1%	707.6	.035	550	10 at 3 kHz	63
21	330.00 ±1%	491.3	.036	216	27 at 3 kHz	59
22	1,000.00 ±1%	1228.0	.019	1400	10 at 3 kHz	70
23	1,000.00 ±1%	852.7	.017	662	16 at 2 kHz	88
24	4,000.00 ±1%	1705.0	.0084	2295	16 at 2 kHz	90

1/ Complete qualification testing and approval to M27/215-24 shall be sufficient to grant qualification approval to M27/215-01 through M27/215-23.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0732)

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